Q.24 Give the composition and properties of abrasive resistive paints.

Q.25 Write short note on matrix materials.

Q.26 Explain the signal sensing materials used in aircrafts.

Q.27 Explain what parameters are taken care of while welding light alloys?

Q.28 Classify various NDT techniques and their applications.

Q.29 Give the safety precautions in use of composite materials.

Q.30 Differentiate between thermoset and thermoplastic with examples.

Q.31 Explain the procedure of thermo-bagging.

Q.32 Explain with diagram the Indian standard angles of any thread.

Q.33 Give the causes of welding defects.

Q.34 Explain methods of corrosion detection.

Q.35 Write the properties and applications of styrofoam.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What is the need of curing of composite materials. Explain the process of it with the help of a neat labelled diagram.

Q.37 Explain various spring materials, their characteristics and their applications.

Q.38 Explain various sensor materials with their applications.

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Roll No. ....

5th Sem / AME

Sub : Aircraft Material & Material Science-II

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following fiber is not used in composite material

a) Glass fiber b) Carbon fiber

c) Aramid fiber d) None of the above

Q.2 In a sandwich composite the core material primarily contributes for

a) Bending stiffness

b) Tensile strength

c) Shear stiffness

d) Compressive strength

Q.3 Fill form corrosion occurs primarily

a) On base metal surfaces

b) Under painted or plated surfaces

c) In high temperature environments

d) In the absence of oxygen

Q.4 Which type of thermo-couple is best suited for measuring the highest temperature range

a) Type K

b) Type J

c) Noble metal thermocouple
d) Base metal thermocouple

Q.5 In gas welding, neutral flame is characterized by
a) Equal oxygen and hydrogen
b) Equal oxygen and acetylene
c) Equal hydrogen and acytelene
d) None of the above

Q.6 Which of the following materials is used in making aircraft windows?
a) Thick glass b) Polycarbonate
c) Graphite d) Plane glass

Q.7 Which of the following can be used as wing covering in an aircraft?
a) Manganese alloy b) Carbon
c) Titanium d) Aluminum alloy

Q.8 Whit worth thread is ____ standard.
a) British b) American
c) Indian d) Metric

Q.9 Magnesium alloys are best suitable for helicopter construction because they are :
a) Highly resistant to corrosion
b) Lighter in weight than aluminum
c) Less expensive than aluminum
d) More rigid than steel

Q.10 18-8 corrosion resistant steel used in aircraft has :
a) 18% nickel and 8% chromium
b) 18% chromium and 8% nickel
c) 18% iron and 8% manganese
d) 18% aluminum and 8% titanium

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 PRT stands for _____.

Q.12 Pressure bag moulding is relatively expensive due to tooling costs comparing to vacuum bag moulding. (True/False)

Q.13 Primary disadvantages of Liquid penetrate test in NDT is that _____.

Q.14 Curing is a method of _______.

Q.15 The purpose of fillers is _____.

Q.16 A stud is different from bolt in manner_____.

Q.17 Define strux.

Q.18 Pre-impregnated materials are used for _____ purpose.

Q.19 The product is made_____ in electroplating (Positive / negative)

Q.20 Name any one locking device.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the method of crack detection with dye penetrant.

Q.22 Differentiate with diagrams between the terms warps & waft.

Q.23 Explain the inspection method of nuts and bolts in air crafts.